# CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

PROJECT REPORT

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## Abstract

This project demonstrates a machine-learning workflow for estimating whether a telecom customer may leave a service (customer churn). It uses Python for data handling and model development, scikit-learn for preprocessing and classification, and Streamlit for an interactive dashboard. The workflow includes data preparation, feature encoding, model training, evaluation, and a form where a user can enter customer details to obtain a churn-risk estimate.

The repository documentation states that the included demo CSV contains synthetic records created for demonstration. Therefore, results from the default dataset must not be presented as real-world business performance. The model's output is an estimate, not a guarantee.

## 1. Introduction

Customer churn occurs when a customer stops using a company's product or service. Understanding churn can help a business investigate customer experience, service issues, pricing concerns, and retention opportunities. Machine learning can identify patterns in customer data and estimate churn risk for an individual customer.

This project is an educational demonstration of an end-to-end classification application. It is not intended to make real customer decisions without further validation.

## 2. Problem Statement

To build an application that uses customer attributes to estimate whether a telecom customer is likely to churn, and to present that estimate through a simple web dashboard.

## 3. Objectives

Prepare customer data for machine learning.

Process numeric and categorical features consistently.

Train a Random Forest classification model.

Evaluate the model using appropriate classification metrics.

Provide an interactive Streamlit interface for entering customer details and viewing a prediction.

Document the project so another user can install and run it.

## 4. Tools and Technologies

Python — core programming language.

Pandas and NumPy — data handling and numerical operations.

scikit-learn — preprocessing, model training, data splitting, and evaluation.

Streamlit — interactive web dashboard.

Matplotlib — charts and visual summaries.

GitHub — source-code hosting and project sharing.

## 5. Project Files

app.py — Streamlit dashboard and customer prediction interface.

train_model.py — model training and evaluation workflow.

download_real_dataset.py — optional script described in the repository for downloading a public sample dataset.

requirements.txt — Python package dependencies.

README.md — project overview and setup instructions.

.gitignore — files and folders excluded from version control.

## 6. Methodology

1. Load the customer churn dataset.

2. Prepare the data, including converting TotalCharges to numeric form and handling missing values.

3. Remove customer identifiers from model features where appropriate.

4. Split the data into training and test sets using an 80:20 split with stratification.

5. Encode categorical columns using one-hot encoding and scale numeric columns as defined by the preprocessing pipeline.

6. Train a Random Forest classifier.

7. Evaluate the model using metrics such as accuracy, balanced accuracy, ROC-AUC, precision, recall, F1-score, classification report, and confusion matrix.

8. Use the trained pipeline in the Streamlit dashboard to estimate churn risk for entered customer details.

## 7. System Workflow

Customer data input → preprocessing → trained Random Forest model → churn-risk estimate → result shown in the Streamlit dashboard.

## 8. Output and Results

The intended output is an interactive dashboard where a user enters customer attributes and selects the prediction action to receive a churn-risk estimate. The repository documentation also describes churn-distribution and contract-comparison charts, plus model evaluation metrics generated by the training script.

Actual accuracy, ROC-AUC, precision, recall, and other numeric results are not included in this report because they must be taken from a real run of train_model.py. Do not add guessed scores. After running the project, insert the actual metrics and a screenshot of the dashboard here.

Dataset note: the repository README says the default data/customer_churn.csv contains synthetic demo records. This means evaluation results demonstrate the workflow only; they do not establish performance on real telecom customers.

## 9. How to Run the Project

Prerequisites: Python 3.11 or 3.12 and Visual Studio Code (or another terminal).

1. Download or clone the GitHub repository and open the Customer-Churn-Prediction folder.

2. Open a terminal in the project folder.

3. Install dependencies: py -m pip install -r requirements.txt

4. Train and evaluate the model: py train_model.py

5. Start the dashboard: py -m streamlit run app.py

6. Open the local URL printed in the terminal, usually http://localhost:8501.

If the py command is unavailable, try python in its place and follow any additional setup instructions shown by the application.

## 10. Limitations

The bundled dataset is synthetic demo data, according to the repository README.

Prediction quality depends on the data and should be reassessed when a different dataset is used.

A model prediction is an estimate and may be incorrect.

The GitHub repository stores source code; it does not automatically host the interactive Streamlit dashboard.

This educational demo should not be used as the sole basis for real customer decisions.

## 11. Future Scope

Evaluate the model using an appropriately sourced real-world dataset.

Compare multiple algorithms and tune model parameters.

Add model explainability to help users understand factors influencing predictions.

Deploy the Streamlit dashboard online and include its live URL in the repository README.

Add screenshots and verified evaluation metrics to the project report.

## 12. Conclusion

This project demonstrates the main stages of a machine-learning classification application: data preparation, preprocessing, model training, evaluation, and an interactive prediction interface. It provides an educational foundation for understanding customer churn prediction. Before presenting numeric performance claims, the training script should be run and actual evaluation results recorded. The synthetic-data limitation should be clearly stated when submitting or demonstrating the project.

## References

Project repository: https://github.com/princi-v/Customer-Churn-Prediction

Python documentation: https://docs.python.org/3/

scikit-learn documentation: https://scikit-learn.org/stable/

Streamlit documentation: https://docs.streamlit.io/